We create chemistry

Safety data sheet

1. Substance/preparation and manufacturer/supplier identification

Comet EC

Use: crop protection product, fungicide

Manufacturer/supplier:
BASF New Zealand Ltd.
5E City Works Depot
77 Cook Street
Auckland Central, Auckland 1010
NEW ZEALAND
Telephone: +64 9 255-4300
Telefax number: +64 9 255-4307

Emergency information:
National Poisons Centre: 0800 764 766
BASF Emergency Advice Number: 0800 944 955 (24 hour advice in an emergency only)
BASF Emergency Advice Number: +61 3 8855 6666 (If calling from outside New Zealand)

2. Hazard identification

Classification of the substance and mixture:
Acute toxicity: Cat. 4 (oral)
Acute toxicity: Cat. 4 (Inhalation - mist)
Skin corrosion/irritation: Cat. 2
Skin sensitization: Cat. 1B
Specific target organ toxicity — single exposure: Cat. 3 (irritating to respiratory system)
Hazardous to the aquatic environment - acute: Cat. 1
Hazardous to the aquatic environment - chronic: Cat. 1

Label elements and precautionary statement:

Pictogram:

Signal Word:
Warning

Hazard Statement:

H315	Causes skin irritation.
H332	Harmful if inhaled.
H302	Harmful if swallowed.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Precautionary Statement:

P101	If medical advice is needed, have product container or label at hand.
P102	Keep out of reach of children.
P103	Read carefully and follow all instructions.

Precautionary Statements (Prevention):

P280	Wear protective gloves.
P271	Use only outdoors or in a well-ventilated area.
P260	Do not breathe mist or vapour.
P261	Avoid breathing mist.
P272	Contaminated work clothing should not be allowed out of the workplace.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P303 + P352	IF ON SKIN (or hair): Wash with plenty of soap and water.
P333 + P311	If skin irritation or rash occurs: Call a POISON CENTER or physician.
P330	Rinse mouth
P391	Collect spillage.
P362 + P364	Take off contaminated clothing and wash it before reuse.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Other hazards which do not result in classification:
See section 12 - Results of PBT and vPvB assessment.
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/information on ingredients

Chemical nature

crop protection product, fungicide, Emulsifiable concentrate (EC)

Hazardous ingredients

Carbamic acid, [2-[[[1-(4-chlorophenyl)-1H-pyrazol-3-yl]oxy]methyl]phenyl]methoxy-, methyl ester

Content (W/W): 23.86 %	Acute Tox.: Cat. 3 (Inhalation - mist)
CAS Number: 175013-18-0	Skin Corr./Irrit.: Cat. 2
	STOT SE: Cat. 3 (irr. to respiratory syst.)
	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 1
	M-factor acute: 100
	M-factor chronic: 10

Hydrocarbons, C10-C13, aromatics, <1% naphthalene

Content (W/W): < 50 %	Asp. Tox.: Cat. 1
CAS Number: 64742-94-5	Aquatic Acute: Cat. 2
	Aquatic Chronic: Cat. 2

Alcohols, C16-18, ethoxylated propoxylated

Content (W/W): < 30 %	Acute Tox.: Cat. 2 (Inhalation - mist)
CAS Number: 68002-96-0	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 3
	M-factor acute: 1

Naphthalene, 2-methyl-

Content (W/W): < 20 %	Acute Tox.: Cat. 4 (oral)
CAS Number: 91-57-6	Aquatic Chronic: Cat. 2

Naphthalene, 1-methyl-

Content (W/W): < 15 %	Asp. Tox.: Cat. 1
CAS Number: 90-12-0	Acute Tox.: Cat. 4 (oral)
	Aquatic Acute: Cat. 2
	Aquatic Chronic: Cat. 2

biphenyl

Content (W/W): < 5 %	Acute Tox.: Cat. 5 (oral)
CAS Number: 92-52-4	Skin Corr./Irrit.: Cat. 2
	Eye Dam./Irrit.: Cat. 2A
	STOT SE: Cat. 3 (irr. to respiratory syst.)
	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 3
	M-factor acute: 1
	M-factor chronic: 1

Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs., calcium salts

Content (W/W): < 5 %	Acute Tox.: Cat. 5 (oral)
CAS Number: 84989-14-0	Acute Tox.: Cat. 5 (dermal)
	Skin Corr./Irrit.: Cat. 2
	Eye Dam./Irrit.: Cat. 1
	Aquatic Acute: Cat. 2
	Aquatic Chronic: Cat. 3

2-ethylhexan-1-ol

Content (W/W): < 5 %	Flam. Liq.: Cat. 4
CAS Number: 104-76-7	Acute Tox.: Cat. 5 (oral)
	Acute Tox.: Cat. 4 (Inhalation - mist)
	Skin Corr./Irrit.: Cat. 2
	Eye Dam./Irrit.: Cat. 2A
	STOT SE: Cat. 3 (irr. to respiratory syst.)
	Aquatic Acute: Cat. 3

toluene

Content (W/W): < 1 %	Asp. Tox.: Cat. 1
CAS Number: 108-88-3	Flam. Liq.: Cat. 2
	Skin Corr./Irrit.: Cat. 2
	Repr.: Cat. 2 (unborn child)
	STOT SE: Cat. 3 (drowsiness and dizziness)
	STOT RE (Central nervous system): Cat. 2
	Aquatic Acute: Cat. 2
	Aquatic Chronic: Cat. 3

4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

Specific hazards:
carbon monoxide, carbon dioxide, hydrogen chloride, nitrogen oxides, sulfur oxides, halogenated compounds
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire.

6. Accidental Release Measures

Personal precautions:
Use personal protective clothing. Avoid contact with the skin, eyes and clothing. Do not breathe vapour/spray.

Environmental precautions:
Do not discharge into drains/surface waters/groundwater. Do not discharge into the subsoil/soil.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Dispose of absorbed material in accordance with regulations. Wear suitable protective equipment.

7. Handling and Storage

Handling

When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift. No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas.

Protection against fire and explosion:
Vapours may form ignitable mixture with air. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 60 Months

Protect from temperatures below: 10 °C
The product can crystallize below the limit temperature.
Protect from temperatures above: 40 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

8. Exposure controls and personal protection

Components with occupational exposure limits

biphenyl, 92-52-4;
TWA value 0.2 ppm (ACGIHTLV)
TWA value 1.3 mg/m3 ; 0.2 ppm (OEL (NZ))

solvent naphtha, 64742-94-5;
TWA value 200 mg/m3 (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Application restricted to conditions in which there are negligible aerosol exposures.
Skin Designation (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Danger of cutaneous absorption

Naphthalene, 1-methyl-, 90-12-0;
TWA value 0.5 ppm (ACGIHTLV)
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption

Naphthalene, 2-methyl-, 91-57-6;
TWA value 0.5 ppm (ACGIHTLV)
Skin Designation (ACGIHTLV)
The substance can be absorbed through the skin.
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN 374) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

9. Physical and Chemical Properties

Form:	liquid	
Colour:	tan	
Odour:	faintly aromatic	
Odour threshold:	Not determined since harmful by inhalation.	
pH value:	approx. 5 - 7 (1 %(m), 20 °C) (as an emulsion)	
pour point:	approx. -5 °C Information applies to the solvent.	
Boiling range:	approx. 244 - 292 °C Information applies to the solvent.	
Flash point:	approx. 104 °C	(ISO 2719)
Evaporation rate:	not applicable	
Flammability (solid/gas):	not highly flammable	
Lower explosion limit:	0.6 %(V) Information applies to the solvent.	
Upper explosion limit:	7 %(V) Information applies to the solvent.	

Ignition temperature:	425 °C The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.	(Directive 92/69/EEC, A.15)
Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.	
Explosion hazard:	Based on the chemical structure there is no indication of explosive properties.	
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.	
Vapour pressure:	approx. 0.003 hPa (20 °C) Information applies to the solvent.	
Density:	approx. 1.05 g/cm3 (20 °C)	
Relative vapour density (air):	Heavier than air., Information based on the main components.	
Solubility in water:	emulsifiable	
Partitioning coefficient n-octanol/water (log Pow):	not applicable	
Viscosity, dynamic:	approx. 64.2 mPa.s (20 °C, 100 1/s) The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.	(OECD 114)
Viscosity, kinematic:	23.7 mm2/s (40 °C) The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.	(OECD 114)

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.

Substances to avoid:
strong bases, strong acids, strong oxidizing agents

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Acute toxicity

Assessment of acute toxicity:
Of moderate toxicity after single ingestion. Virtually nontoxic after a single skin contact. Of moderate toxicity after short-term inhalation. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
LD50 rat (oral): 317 mg/kg

LC50 rat (by inhalation): > 1.14 - < 5.3 mg/l 4 h (OECD Guideline 403)
An aerosol was tested.

LD50 rat (dermal): > 4,000 mg/kg (OECD Guideline 402)
No mortality was observed.

Irritation

Assessment of irritating effects:
Skin contact causes irritation. Not irritating to the eyes. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
Skin corrosion/irritation rabbit: (OECD Guideline 404)

Serious eye damage/irritation rabbit: (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Sensitization after skin contact possible.

Experimental/calculated data:
Guinea pig maximization test guinea pig:

Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of various animal studies gave no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: toluene
Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Specific target organ toxicity (single exposure):

Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Pyraclostrobin
Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation. The substance may cause damage to the olfactory epithelium after repeated inhalation.

Information on: biphenyl
Assessment of repeated dose toxicity:
The substance may cause damage to the kidney after repeated ingestion of high doses, as shown in animal studies.

Information on: toluene
Assessment of repeated dose toxicity:
Repeated exposure to large quantities may affect certain organs. Damages the central nerve system. The substance may cause deafness after repeated inhalation.

Information on: Benzenesulfonic acid, 4-C10-13-sec-alkyl derivs., calcium salts
Assessment of repeated dose toxicity:

After repeated exposure the prominent effect is local irritation. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

Misuse can be harmful to health.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Toxicity to fish:
LC50 (96 h) 0.0229 mg/l, Oncorhynchus mykiss (EPA 72-1, static)

Aquatic invertebrates:
EC50 (48 h) 0.046 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)

Aquatic plants:
EC50 (72 h) 0.69 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)

EC10 (72 h) 0.11 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Chronic toxicity to fish:
No observed effect concentration (98 d) approx. 0.00235 mg/l, Oncorhynchus mykiss (OECD Guideline 210, Flow through.)

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), 0.004 mg/l, Daphnia magna (OECD Guideline 202, part 2, semistatic)
The details of the toxic effect relate to the nominal concentration.

No observed effect concentration (28 d), 0.00128 mg/l, Mysidopsis bahia (OPP 72-4 (EPA-Guideline), Flow through.)
The statement of the toxic effect relates to the analytically determined concentration.

Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment transport between environmental compartments:
Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
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Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Bioaccumulation potential:
Bioconcentration factor: 379 - 507, Oncorhynchus mykiss (OECD-Guideline 305)
Accumulation in organisms is not to be expected.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Domestic transport:

Packing group:	III
ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains PYRACLOSTROBIN, SOLVENT NAPHTHA)

Further information
Hazchem Code:3Z
IERG Number:47

Sea transport
IMDG

Packing group:	III
ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Marine pollutant:	YES
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains PYRACLOSTROBIN, SOLVENT NAPHTHA)

Air transport
IATA/ICAO

Packing group:	III
ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains PYRACLOSTROBIN, SOLVENT NAPHTHA)

Further information
Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2).

15. Regulatory Information

Hazard determining component(s) for labelling: PYRACLOSTROBIN, SOLVENT NAPHTHA, ALCOHOLS, ETHOXYLATED, PROPOXYLATED

Other regulations

To avoid risks to man and the environment, comply with the instructions for use.

APVMA Approval Number P006017

A certified handler is not required for the handling of this substance.
Tracking requirements do not apply to this substance.

16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.